Lecture 5: What Is It Like to Be a Baby: The Development of Thought

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-5 (CC BY-NC-SA 3.0).

Before you start

This lecture is about cognitive development: how children come to understand the world. Professor Bloom presents Piaget's stage theory, and then modern studies suggesting that babies know more than Piaget thought. He ends with the social world, autism and the idea that the mind may have separate "modules."

Learning goals

After this lecture you should be able to:

- Describe Piaget's stages and his main findings.
- Explain how researchers use looking time to study babies.
- Give two examples of studies suggesting that babies know more than Piaget thought.
- Compare three explanations of development: brain growth, inhibition and learning.
- Explain the false-belief task and what it shows about autism.

Key ideas

1. Jean Piaget, Stage Theory and Its Limits

- Piaget saw children as active thinkers, "little scientists" who build small theories of the world (**schemas**).
- Two mechanisms drive learning: **assimilation** (using what you do on new things, like a baby who sucks a breast and then sucks a bottle) and **accommodation** (changing how you do it, like shaping the mouth differently).
- **Sensorimotor stage:** the child touches and sees but does not reason. Slowly the child gets **object permanence**, knowing that things exist when out of sight. Piaget said that before six months, a hidden toy is simply "gone" for the baby. Nine-month-olds make the **A-not-B error**: after finding a toy at A several times, they still reach for A after seeing it moved to B.
- **Preoperational stage:** the child can represent the world but is **egocentric**: he cannot understand that others see things differently (the three mountains task). Children also fail **conservation**: they think a row of checkers that is spread out has more checkers.
- **Concrete operations** (seven to twelve): children solve conservation but cannot reason abstractly. At around twelve, abstract and scientific reasoning begins.

- Piaget did better than Freud or Skinner: his claims were falsifiable, and his findings were striking. But he relied on questions and answers, and young children are not good with language, so he may have underestimated what they know.

2. The Modern Science of Infant Cognition

- Babies cannot talk, so researchers use clever methods: brain waves, sucking on a pacifier and, most of all, **looking time**. Elizabeth Spelke has done the most research with this method.
- Two looking methods: **preference** (which picture does the baby look at?) and **habituation and surprise**. A baby who is bored by green patches will look longer at a red one if it looks different.
- Spelke and Baillargeon: babies expect a bar moving behind a block to be one complete bar, and look longer at a broken bar.
- Rene Baillargeon: six-month-olds look longer when a screen keeps moving through the place where a hidden block is.
- Karen Wynn's lab at Yale: six-month-olds expect 2 - 1 = 1, and look longer when 2 - 1 = 2.
- Some things develop later: six-month-olds are surprised by a block floating in mid-air, but not by a badly placed block; twelve-month-olds expect that block to fall.
- Three explanations of development: (1) **neural maturation**: pruning of neurons, fast growth of connections (peaking at about two years) and myelination, which is not finished even in teenagers' frontal lobes; (2) **inhibition**: Adele Diamond found that children making the A-not-B error reach for A but look at B; (3) **learning**: Michael McCloskey found that many college students wrongly think a ball shot through a tube keeps curving, but almost nobody makes this error with water.

3. Babies in the Social World

- Newborns prefer to look at a pattern that looks like a face. Andrew Meltzoff found that newborns often stick out their tongues when an adult does.
- Professor Bloom, Valerie Kuhlmeier and Karen Wynn showed nine- and twelve-month-olds movies in which one character helps a ball reach its goal and another hinders it. Babies looked longer when the ball approached the hinderer. This is "preliminary evidence" that they understand helping and hindering.
- Kiley Hamlin began studies to see which character babies reach for. In such studies, the person offering the choice is **blind**: she does not know what the baby saw, so she cannot influence the answer.
- This evidence is "tentative, very controversial." And young children can also be "very clueless" about people, as films of deception tasks showed.

4. Question and Answer on Learning and Development

On the tasks in the films, three-year-olds and young four-year-olds usually fail, and children succeed around age four or five. Operant conditioning works less well with babies than looking-time methods.

5. Review of Studies Presented in Class; Autism

- Why do young children fail deception tasks? Maybe they do not understand minds. Or maybe they have problems with **inhibition**: a boy kept pointing at the sticker he wanted, though a "mean

monkey" then took it. Even adults are slower at lying tasks.

- Piaget believed in general changes in all thinking. Noam Chomsky and Jerry Fodor suggest instead separate **modules**: pre-wired systems for reasoning about different areas, such as objects and people.
- Evidence: **autism**, which affects about one in a thousand people, mostly boys (the 2007 lecture's figure; current estimates are much higher). People with autism may have no problems with the physical world or math, but serious problems with people. Simon Baron-Cohen calls this "mind blindness."
- In Baron-Cohen's "Charlie" task, autistic children often cannot say which chocolate Charlie will take, though he is looking at it.
- The **false-belief task:** Maxie puts a ball in the cupboard and leaves. Another doll moves it under the bed. Where will Maxie look? The right answer is the cupboard, because Maxie has a false belief. Four- and five-year-olds pass; two- and three-year-olds find it difficult; many people with autism fail.

6. Question and Answer on Autism

Professor Bloom said that autism is not simply a developmental delay, that most people with autism have severe problems, and that special "savant" skills, as in *Rain Man*, are very rare. He closed with Stephen Jay Gould's point that development can answer questions about adults, and D'Arcy Thompson's line: "Everything is the way it is because it got that way."

Key vocabulary

Term	Meaning in simple English	Example from the lecture
assimilation	Using an old way of acting on new things	Sucking a bottle like a breast
accommodation	Changing your way of acting to fit something new	Shaping the mouth differently
object permanence	Knowing that things exist when you cannot see them	Peek-a-boo surprises young babies
A-not-B error	Searching where an object was before, not where it is now	The nine-month-old still reaches for A
egocentric	(Piaget) unable to see the world from another person's view	The three mountains task
conservation	Knowing an amount stays the same when its shape changes	Spreading out a row of checkers
looking time	How long a baby looks at something; longer means surprise or interest	Babies look longer at 2 - 1 = 2
inhibition	The ability to stop yourself from doing something	Reaching for A while looking at B

Term	Meaning in simple English	Example from the lecture
blind (in research)	Not knowing what the participant experienced	The person offering the baby a choice
module	A separate built-in system for one kind of thinking	An object module and a social module
false belief	A belief that does not match reality	Maxie thinks the ball is in the cupboard

Check your understanding

1. What is the difference between assimilation and accommodation? Use Piaget's example.
2. What is object permanence, and what did Piaget claim about babies under six months?
3. A five-year-old says a spread-out row of checkers has more checkers than a short row with the same number. What concept is the child missing?
4. How does the habituation method work? Use the green and red example.
5. Describe the study from Karen Wynn's lab. What does it suggest?
6. Give two possible explanations of the A-not-B error.
7. Describe the false-belief task. Which children usually pass it?
8. How is autism used as evidence for separate modules in the mind?

Answer guide

1. Assimilation is using your way of acting on new things (sucking a bottle). Accommodation is changing how you act (shaping your mouth differently).
2. It is knowing that things still exist when you cannot see them. Piaget said babies under six months think a hidden object is just gone.
3. Conservation: the number stays the same when the row is spread out.
4. Show a green patch again and again until the baby gets bored, then a red patch. If the baby sees it as different, it looks longer.
5. Six-month-olds look longer when 2 - 1 = 2 than when 2 - 1 = 1. They seem to know simple arithmetic and expect hidden objects to still exist.
6. Babies may not understand objects. Or they cannot inhibit (stop) reaching to the old place; Adele Diamond found they look at B while reaching for A.
7. Maxie leaves a ball in the cupboard; it is moved while he is away. The child must say Maxie will look in the cupboard. Four- and five-year-olds usually pass.
8. People with autism can do well with objects and math but have serious problems with people. So social thinking may be a separate system.

English practice

Useful phrases from the lecture

- "It's possible that..." - presents a possible idea. Example: "It's possible that... he systematically underestimated what children and babies know."
- "Another possibility is..." - gives a different explanation. Example: "Another possibility is there's problems with inhibition."
- "This suggests that..." - carefully says what evidence shows. Example: "So, this suggests that some social understanding may be there from the very start."
- "Often 90% of the game is..." - says what matters most in an activity. Example: "Often 90% of the game is discovering a clever method through which to test your hypotheses."

Speaking task (with your AI tutor)

- Explain how scientists study what babies know in 4-6 sentences. Use these terms: *looking time*, *habituation*, *surprise*.
- Explain the false-belief task and its link to autism in 4-6 sentences. Use these terms: *false belief*, *mind blindness*, *module*.

Writing task

- Write one paragraph (80-120 words): Describe one modern study that challenges Piaget's view of object permanence, and explain why his methods may have underestimated what babies know.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 5. Ask me questions about Piaget's stages, one at a time."
- "Quiz me on the key vocabulary of Lecture 5."
- "I will explain the A-not-B error and two possible reasons for it. Check my understanding and my English."

Readings from the original course

- Gray, Peter. Psychology (5th edition), chapter 11
- Gopnik, Alison, Patricia Kuhl, and Andrew Meltzoff. "The Scientist in the Crib: Minds, Brains, and How Children Learn." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 146-153

My notes